CARECHRONICLE

Design Approach

UX, Visual System, Product Architecture & Design Diagrams

Visual direction: Premium, calm, trustworthy — with a distinctive pink-led identity.

Document	Design Approach v2.0
Product	CareChronicle
Scope	Product design + UX + UI + architecture visualization
Primary use case	Patient-owned medical record & care organization
Initial focus	Cancer / complex longitudinal care
Design palette	Pink-led, high-contrast, restrained accent usage

Organize the records. Preserve the story. Simplify the journey.

1. Design North Star

CareChronicle should feel like a calm, trustworthy medical filing system — not a flashy chatbot.

The interface should transform a messy collection of reports into a coherent journey without pretending that the system is a clinician. The product earns trust through preservation, traceability, predictable organization and controlled AI assistance.

> ***North-star experience:** "I can open CareChronicle and quickly understand where my records are, what happened recently, what changed, and what I should take to the next appointment."*

Design success is measured by reduced searching, reduced reconstruction of history, faster appointment preparation and confidence that the original evidence remains available.

2. Core Design Principles

Principle	Design approach
Trust before automation	AI suggests; people verify important relationships and metadata.
Original evidence first	Every derived insight can lead back to the original document.
Timeline over file pile	The journey is the mental model; files support each event.
Appointment as a core unit	A visit groups reports, prescriptions, notes, bills and instructions.
Show uncertainty	Use explicit review states instead of silent assumptions.
Calm visual hierarchy	Use whitespace and progressive disclosure; avoid dashboard overload.
AI is the engine	AI appears inside useful workflows rather than becoming the product identity.
Patient control	Sharing, access, deletion and export are explicit and understandable.
Messy reality is normal	Design for scans, photos, duplicates, conflicting dates and incomplete extraction.

3. Visual Design Direction — Pink System

Pink is the primary brand and interaction color. It should communicate warmth and confidence without making the interface feel playful or cosmetic. The safest approach is a **deep pink primary + pale pink surfaces + neutral backgrounds + restrained semantic colors**.

Core palette

Token	Suggested value	Usage
Primary Pink	#C2185B	Primary actions, active navigation, key brand elements
Dark Pink	#8E1748	Headings, strong emphasis, cover/title treatment
Accent Pink	#E91E63	Secondary interactive emphasis, selected states
Light Pink	#FCE4EC	Callouts, selected surfaces, subtle backgrounds
Soft Pink	#F8BBD0	Borders, dividers, low-emphasis highlights
Neutral Surface	#FFF7FA	Very light tinted app surfaces
Text	#26202A	Primary text
Muted Text	#6B6570	Secondary metadata
Success	#287A57	Confirmed/complete states only

Token	Suggested value	Usage
Warning	#9A6700	Needs review/uncertainty
Error	#A33A3A	Actual errors or destructive warnings

Important: semantic colors must not replace text labels. Pink is the product identity; green, amber and red remain functional status colors.

4. Information Architecture

Area	Purpose	Primary actions
Home	Current context and next action	Add, review, prepare, search
Timeline	Longitudinal care story	Explore events, open evidence
Medical Vault	Complete document repository	Search, filter, preview, link
Appointments	Visit-centered organization	Prepare, review, add, export
Packages	Purpose-built outputs	Doctor, second opinion, insurance
Profile & Access	Identity, caregivers, permissions and privacy	Manage access, sharing, controls

5. System Architecture — Design View

The architecture should reinforce the product philosophy: preserve originals, derive structured information, connect it to care events, and present useful workflows through a patient-facing experience.

Patient / Caregiver
Mobile App + Web

→

Experience Layer
Home · Timeline · Vault · Appointments · Packages

→

Application Services
Auth · Records · Search · Sharing · Exports

Document Intelligence
OCR · Classification · Metadata · Entity Extraction · Relationship Suggestions

↕

Core Medical Data
Patients · Care Events · Appointments · Documents · Treatments · Permissions

Storage
Encrypted object storage + database + backups + audit log

↕

External / Future Integrations
EHR · Labs · Hospital portals · Insurance workflows

Architecture principle: the AI/document-intelligence layer should never become the system of record. The source document and verified medical-record entities remain the authoritative foundation.

Design implications

- The UI can show AI processing as a supporting state, but the user's records remain accessible even if AI processing fails.
- Core navigation should work from structured records and documents; it should not require a chat session.
- Future integrations should feed the same care-event/document model rather than create separate silos.
- Exports should be generated from structured data + selected originals, with a manifest of included documents.

6. Core Data Relationship Diagram

The design system should be based on explicit relationships rather than folders alone. This allows the same document to be discoverable through the Vault, appointment, timeline and package workflows.

Patient / Case
1 patient → 1+ care journeys

1 : N

Care Event
Appointment · Admission · Test · Treatment

1 : N

Document
Report · Prescription · Bill · Note

supp
orts

Package
Doctor summary · Second opinion · Insurance

Object	Design meaning
Patient / Case	The ownership and context boundary for the medical journey.
Care Event	An appointment, admission, test, treatment or other meaningful event.
Document	The preserved source artifact and its metadata.
Package	A curated output assembled for a specific purpose.

7. Upload & Document Intelligence Flow

Upload is the first major trust interaction. The user should not need to understand OCR, classification or metadata schemas.

1. Upload PDF · Image · Scan → **2. Preserve** Original saved first → **3. AI Process** OCR · classify · extract → **4. Review** Confirm uncertain fields → **5. Organize** Vault · Timeline · Appointment

Recommended interaction rule

- **Save first, enrich second:** the original is preserved before the system asks for review.
- **Ask only when it matters:** low-confidence or consequential associations should be confirmed.
- **Never block on perfection:** an unlinked document can remain safely in the Vault.
- **Make correction easy:** wrong classification or date should be fixable in seconds.

> Metadata should distinguish at least document date, appointment date and billing date. These dates should not be forced into a single universal timestamp.

8. Appointment-Centered Experience

Appointments are the bridge between raw records and real-world care. A patient should be able to open one visit and see the evidence and actions surrounding it.

Before visit Review recent records → **What changed?** Compare selected evidence → **Prepare** Questions + key records → **Package** Summary / share / export

Appointment area	Design content
Header	Date, provider/specialty, status and context.
Since last visit	New records, changes, new prescriptions and relevant events.
What changed?	Evidence-based comparison with source links.
Questions / notes	Patient-entered notes and questions, clearly separated from clinician records.
Documents	Reports, prescriptions, notes, bills, receipts and instructions.
Actions	Generate summary, curate package, share/export.

9. “What Changed?” UX Architecture

This is a signature workflow. It should present recorded differences rather than diagnose or interpret them.

Step	UI behavior
1. Choose comparison	Select latest vs previous report/event or use the suggested pair.
2. Show extracted differences	Previous value → latest value, document section and date where reliable.
3. Attach provenance	Every meaningful difference links to the source page/document.
4. Flag uncertainty	If extraction is incomplete or ambiguous, say so clearly.

Step	UI behavior
5. Optional generated summary	Plain-language synthesis labeled as AI-generated and evidence-linked.

Safety boundary: the interface can say what records show and how records differ. It should not convert those differences into diagnosis, prognosis or treatment advice.

10. Doctor / Appointment Preparation Design

The output should be concise enough for a real consultation while retaining access to the full record.

- **Since last visit:** the most relevant new events and records.
- **Key records:** curated evidence with one-tap originals.
- **Questions:** patient-entered items.
- **Summary:** structured one-page output.
- **Share/export:** explicit package contents before sending.

> *Generated summaries should be evidence-linked. The design should make it possible to move from a summary statement to the underlying source record.*

11. Action Packages — Second Opinion & Insurance

Package	Design objective	Core interaction
Second opinion	Create a professional, curated clinical record packet.	Select care period → review records → preview → export/share
Insurance	Organize potentially relevant medical/financial documentation.	Select episode → gather docs → review what is present/not found → export
Doctor summary	Give a clinician a concise longitudinal context.	Select visit → review changes → generate → verify → share

Insurance UX must never imply that CareChronicle knows an insurer's exact requirements unless those requirements have been explicitly supplied/verified. Prefer language such as “not found in CareChronicle” rather than “missing required document.”

12. AI Trust & Provenance System

Information state	Meaning	UI treatment
Original	Direct source artifact	Source document action
Extracted	Machine-read value	Extracted label + source
User-entered	Added or corrected by user	Added by you
Suggested	AI-inferred classification/relationship	Suggested + review action
Generated	AI-created summary text	AI-generated + evidence links

Visual hierarchy rule: generated text must never look more authoritative than the source document.

13. Edge-Case & Failure-State Design

Situation	Design response
Poor scan	Preserve original; show extraction quality issue; allow retry/manual metadata.

Situation	Design response
Handwriting	Do not imply reliable extraction; keep original and allow manual classification.
Duplicate	Suggest possible duplicate; user decides whether to keep.
Conflicting dates	Show document/appointment/billing dates separately.
No appointment match	Keep in Vault; allow later linking.
Wrong AI classification	Fast correction; corrected value becomes the trusted user-confirmed value.
Incomplete extraction	Expose uncertainty instead of filling gaps silently.
Multiple cases	Explicit case context before association to prevent cross-case filing.

14. Screen-by-Screen Design Approach

Screen	Primary job	Key elements
Onboarding	Establish ownership/context	Patient/case setup, privacy explanation, first upload
Home	Orient user quickly	Next appointment, recent care, review queue, quick actions
Upload	Add evidence with low friction	Picker/camera, progress, original preservation
AI Review	Confirm important machine suggestions	Suggested type/date/provider/appointment + edit
Vault	Retrieve any record	Search, filters, document cards, relationship context
Document Viewer	Inspect source evidence	Full document, metadata, linked event, share
Timeline	Understand the journey	Care events, milestones, supporting documents
Appointments	Prepare around visits	Visit list, preparation state, linked records
Appointment Workspace	Manage one care event	Since-last-visit, documents, notes, package actions
Packages	Create useful outputs	Doctor, second opinion, insurance package builder
Access & Privacy	Maintain control	Caregivers, permissions, sharing history, data controls

15. Design System Components & States

Component	Required states
Document card	Default · Processing · Needs review · Verified · Error · Shared
Appointment card	Upcoming · Completed · Incomplete · Archived
Timeline event	Collapsed · Expanded · Selected · Missing evidence
AI suggestion	Suggested · Accepted · Rejected · Edited
Upload	Idle · Uploading · Processing · Success · Partial failure
Package builder	Selected · Omitted · Uncertain · Ready to export
Permission	View · Contribute · Export/share · Revoked

16. Mobile, Web & Print Strategy

Context	Design priority
Mobile	Upload, appointment prep, search, timeline, quick access, sharing.
Desktop/Web	Bulk document review, side-by-side source inspection, package building and administration.
PDF/Print	Professional summaries/packages with clear dates, provenance and document manifest.

Desktop should not simply be a stretched mobile UI. Use larger space for document preview, metadata review and package curation.

17. Accessibility & Localization

- Target WCAG 2.2 AA as the design baseline where applicable.
- Never communicate status through color alone.
- Maintain comfortable touch targets and scalable text.

- Use plain-language action labels.
- Support screen readers with meaningful document/status labels.
- Design for older users and users under stress.
- Support multilingual content inside documents without destroying the original.
- If translation is offered, label it as translation/generated content and preserve the source.

18. Privacy & Sharing UX

Privacy is part of the primary experience. A patient should always know what is being shared, with whom and for what scope.

- Show current access to a case.
- Make package contents visible before sharing.
- Use explicit recipient and scope confirmation.
- Support revocation/expiry where implemented.
- Show sharing history.
- Make deletion/export controls discoverable.
- Explain AI processing in plain language.
- Avoid public-by-default sharing.

The key trust question is: **“Did I just share my entire medical history?”** The interface must make the answer unambiguous.

19. Design Validation & Testing

Test	Question	Success signal
Card sort	Do users understand the organization model?	Strong agreement on core categories
Upload prototype	Do users trust AI suggestions?	Users can verify/correct without confusion
Retrieval test	Can users find a requested record quickly?	Important record found without manual file hunting
Appointment test	Does the appointment model make sense?	User prepares a visit without instruction
Comparison test	Is “What Changed?” understandable and safe?	Users identify recorded differences and sources
Sharing test	Is package scope clear?	No ambiguity about recipient/content

20. Design Metrics

Metric	Definition
Time to organized record	Upload to useful placement in Vault/timeline/appointment.
Retrieval time	Time to locate a requested record.
AI correction rate	How often suggestions require user correction.
Appointment preparation completion	Eligible visits for which preparation is completed.
Source traceability	Generated summary items traceable to source evidence.
Sharing comprehension	Users understand what a package contains before sharing.

Metric	Definition
Trust score	User confidence in preservation and organization.

21. Final Design Direction

CareChronicle should not try to look like the smartest healthcare app. It should feel like the **most dependable place to organize and retrieve a patient's medical story.**

The pink visual identity should make the product distinctive while the underlying UX remains restrained, professional and evidence-first.

Every major interaction should reinforce three ideas:

- **Your records are preserved.**
- **Your journey is organized.**
- **You remain in control.**

The strongest product moment is not “Ask AI about my report.” It is: **“Everything related to my care is here, I can see what happened, I can see what changed, and I can take the right records to the next person who needs them.”**

Design mantra: ***Organize the records. Preserve the story. Simplify the journey.***

This document should be used with the CareChronicle PRD as the UX/UI and architecture-visualization foundation for the design and engineering teams.